BlueSky experiment: two aircraft types in trail on final approach (Kano)

Muhammad Ibrahim Zubair | note accompanying my application | simulation only, simplified model

Question

When a faster aircraft follows a slower one on the same final approach, how fast does the separation erode, and what does a speed instruction do to it?

Set-up

Two aircraft on the runway 06 extended centreline at Kano (about 12.05 N, 8.52 E, approximate position of the airport; track 060), level at 5000 ft, no wind. Leader APK201 (Embraer 145, 130 kt CAS) starts 25 NM from the airport; follower APK305 (Boeing 737-800, 160 kt CAS) starts 8 NM behind it. Baseline: no instruction. Sequenced: speed instruction to 130 kt for the follower at minute 3. Both runs stop at 6 minutes.

Results

Run	Follower distance to airport	Separation at 6 min	Hand calculation
Baseline (no instruction)	15.8 NM	**4.8 NM** (below 5 NM)	4.8 NM
Sequenced (instruction at minute 3)	16.8 NM	**5.8 NM**	6.4 NM (instant speed change assumed)

Separations were computed from the latitude and longitude printed by BlueSky (rounded to one arc-second, about 0.03 NM). The hand calculation used true airspeed about 7.7 % above calibrated speed at 5000 ft (140 kt and 172 kt), a closing speed of about 32 kt.

Discussion

- Without intervention, the 30 kt difference reduced the 8 NM gap to 4.8 NM in six minutes, as expected from gap change = speed difference x time.
- The instruction kept the separation above 5 NM, but the result (5.8 NM) is lower than the 6.4 NM of the instant-change calculation. The follower covered about 1.0 NM less than in the baseline, which suggests that it took roughly two minutes to slow from 160 to 130 kt (an estimate from the positions; I did not check BlueSky's deceleration model). Until the speeds matched, the gap kept shrinking.
- Consequence: a speed instruction has to be given early, and the time it needs depends on the aircraft type. For a mixed fleet with very different performance, the time and distance needed for such corrections differ by type, which affects safety, capacity and controller workload.

Limitations and next steps

Level flight, constant tracks, no wind, no descent, no wake-turbulence minima, two aircraft only, and a single run per case. My BlueSky experience is basic. Next steps: log positions automatically, vary the speed difference, the instruction time and the aircraft types, add a descent profile and wake separation, and define metrics (minimum separation, time to loss of separation, number of instructions, throughput) over many scenarios.

Appendix: scenario files

```
# baseline.scn
0:00:00.00>CRE APK201 E145 11.8392 8.1563 60 5000 130
0:00:00.00>CRE APK305 B738 11.7724 8.0385 60 5000 160
0:00:00.00>TRAILS ON
0:00:01.00>FF
0:06:00.00>HOLD
# sequenced.scn: same, with one extra line before HOLD:
0:03:00.00>SPD APK305 130
```